FIITJEE
ALL INDIA TEST SERIES
JEE (Advanced)-2023
CONCEPT RECAPITULATION TEST – IV
PAPER –1
TEST DATE: 26-04-2023

ANSWERS, HINTS & SOLUTIONS

Physics

PART – I

Section – A

1. BCD

Sol. Let at time t the angle between magnetic field and area vector (semicircle) be θ, then $\theta = \omega t$

$$\phi = \vec{B}\cdot\vec{S} = \frac{\pi a^2 B}{2}\cos\omega t.$$

$$\varepsilon = -\frac{d\phi}{dt} = \frac{\pi B a^2 \omega}{2}\sin\omega t$$

$$\varepsilon_0 = \frac{\pi B a^2}{2\sqrt{LC}} \text{ peak emf}$$

Since the circuit is in resonance,

$$|z| = R \quad \Rightarrow \quad i_0 = \frac{\pi B a^2}{2R\sqrt{LC}} \text{ peak current}$$

$$i_{rms} = \frac{i_0}{\sqrt{2}} \quad \Rightarrow \quad i_{rms} = \frac{\pi B a^2}{2R\sqrt{2LC}}$$

$U_C = \frac{1}{2}CV_0^2 \rightarrow$ max energy, $V_0 \rightarrow$ peak voltage

$$V_0 = i_0 X_c = \frac{i_0}{C\omega} = \frac{i_0\sqrt{LC}}{C}$$

$$U_C = \frac{1}{2}C\times\frac{\pi^2 B^2 a^4}{4R^2C^2} = \frac{\pi^2 B^2 A^4}{8R^2 C}$$

$$P_{Ext} = P_{Dissipated} = \varepsilon_0 i_0 = \frac{\pi B a^2}{2\sqrt{LC}}\times\frac{\pi B a^2}{2R\sqrt{LC}}, \qquad P_{Ext.} = \frac{\pi^2 B^2 a^4}{4LCR}$$

2. ABC

Sol. $I = \frac{1.06z^2}{n^3}$ mA

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For H atom $z = 1$ and first orbit $n = 1$

$I = 1.06$ mA. So, choice (A) is correct.

Magnetic field $B = \dfrac{12.5z^3}{n^5}$ Tesla

$B = 12.5$ Tesla. So, choice (B) is correct.

$\Delta E = 13.6z^2\left[1-\dfrac{1}{4}\right]$

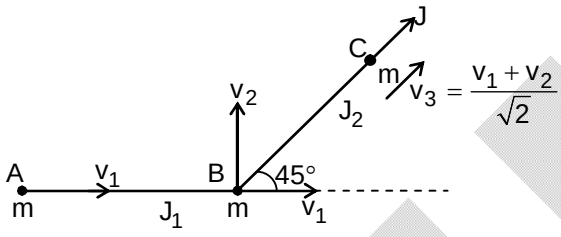
$= 13.6 \times \dfrac{3}{4} = 10.2$ eV. So, choice (C) is correct.

3. ABC

Sol. $\dfrac{J_2}{\sqrt{2}} = 2mv_1$

$\dfrac{J_2}{\sqrt{2}} = mv_2$

$\Rightarrow \quad v_2 = 2v_1$



For C,

$J - J_2 = \dfrac{m}{\sqrt{2}}(v_1 + v_2)$

$J - 2\sqrt{2}mv_1 = \dfrac{3mv_1}{\sqrt{2}} \quad \Rightarrow J = \dfrac{7mv_1}{\sqrt{2}} \quad \Rightarrow v_1 = \dfrac{\sqrt{2}J}{7m}$

$v_2 = \dfrac{2\sqrt{2}J}{7m}$

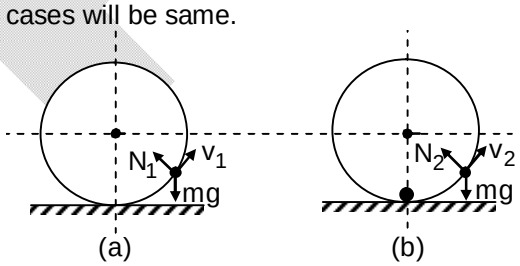
$V_A = V_1 = \dfrac{\sqrt{2}J}{7m} \qquad V_B = \dfrac{\sqrt{10}J}{7m}$

$V_c = \dfrac{V_1 + V_2}{\sqrt{2}} = \dfrac{3J}{7m}$.

4. A

Sol. F.B.D in two cases is as shown.

In case (b) normal will be less and in case (b) work done by magnetic field is zero so height in two cases will be same.



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5. ACD

Sol. Power of gravitational force by A on B is $FV\cos\theta \neq 0$ as $\theta \neq 90°$

$V_{AB} = \sqrt{3}V$

$\text{Power} = FV_{AB}\cos 60°$

$= \dfrac{\sqrt{3}mv^3}{2R}$

6. BD

Sol. Cable in electrically neutral

$\therefore \quad \sigma \times 2\pi b\ell + \rho\pi a^2\ell = 0$

$\dfrac{\sigma}{a^2} + \dfrac{\rho}{2b} = 0$

For $r < a$

$E \times 2\pi r\ell = \dfrac{\rho\pi r^2\ell}{\epsilon_0}$

$E = \dfrac{\rho r}{2\epsilon_0}$

For $a < r < b$

$E \times 2\pi r\ell = \dfrac{\rho\pi a^2\ell}{\epsilon_0}$

$E = \dfrac{\rho a^2}{2\epsilon_0 r}$

For $r > b$

$E \cdot 2\pi r\ell = 0$

$\therefore \quad E = 0$

7. C

Sol. At free end lateral inversion takes place and at fixed end both lateral and transverse inversion take place.

8. C

Sol. $V = \omega A$

Where A is distance of equilibrium position from relaxed position of spring.

(I) $A = \dfrac{mg}{k} = 0.1$ & $\omega = \sqrt{\dfrac{k}{m}} = 10$

(II) $A = \dfrac{mg}{4k} = \dfrac{0.1}{4}$ & $\omega = \sqrt{\dfrac{4k}{m}} = 20$

(III) $A = \dfrac{mg}{k} = 0.1$ & $\omega = \sqrt{\dfrac{k}{m}} = 10$

(IV) $A = \dfrac{4mg}{k} = 0.4$ & $\omega = \sqrt{\dfrac{k}{4m}} = 5$

9. D

Sol. Only in case (III) charge induced on the inner surface is non-uniform. Charge on inner and outer surface of conductor in each case are equal and opposite.

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10. B

Sol. $W_N = \vec{N}.\vec{s}_w + \vec{N}.\vec{s}_{bw}$

$= \vec{N}.\vec{v}_w t + 0$

$= \vec{N}.\vec{v}_w \sqrt{\frac{2\sqrt{2}.h}{g}}$

Section – B

11. 0.50

Sol. $(F_{net}) = \mu_k (2m) g = (m_{total}) a_{cm}$

$a_{cm} = \mu_k g$

$S_{cm} = 0 + \frac{1}{2}(\mu_k g) t^2$

$S_{cm} = \frac{m_1 S_1 + m_2 S_2}{m_1 + m_2}$

$-\frac{1}{2}(\mu_k g) t^2 = \frac{(m)(x) + (m)(x - \ell)}{m + m}$

$x = \frac{\ell - (\mu_k g) t^2}{2} = 0.50 \text{ m}$

12. 4.70

Sol. $\lambda = 2R = 2R_0 A^{1/3}$

$P = \frac{h}{\lambda} \quad \Rightarrow \quad E = \frac{P^2}{2m} = \frac{h^2}{2m\left(4R_0^2 A^{2/3}\right)}$

$E = \frac{\left(6.62 \times 10^{-34}\right)^2}{2 \times 1.67 \times 10^{-27} \times 4\left(1.3 \times 10^{-15}\right)^2 (128)^{2/3}}$ Joule

$= 4.72$ MeV.

13. 0.75

Sol. x_0 = initial compression

x = final elongation

$x = \frac{\rho g h A + mg}{k} = \frac{1}{2}$

$\frac{1}{2} k x_0^2 = \frac{1}{2} k x^2 + mg(x + x_0)$

$x_0 = 1 \text{ m}$

Maximum compression from equilibrium compression

$x' = x_0 - \frac{mg}{k}$

$= 0.75$ m

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14. 500.00

Sol. $\left[P_{in_1}\right]$ P_1 $\quad P_{in_1} - P_1 = \dfrac{4T}{r}$

$\left[P_{in_2}\right]$ p_2 $\quad P_{in_2} - P_2 = \dfrac{4T}{2r} = \dfrac{2T}{r}$

$$P_{in_1} \times \frac{4}{3}\pi r^3 = P_{in_2} \times \frac{4}{3}\pi(2r)^3$$

$$P_{in_1} = P_{in_2} \times 8$$

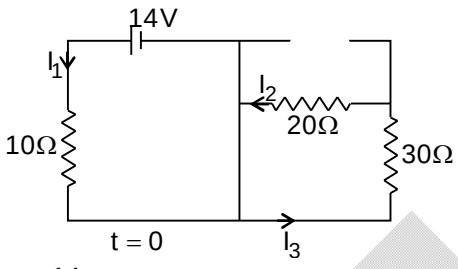
$$P_1 + \frac{4T}{r} = \left(\frac{2T}{r} + P_2\right)8$$

$$\frac{P_1}{8} + \frac{T}{2r} - \frac{2T}{r} = P_2 \quad : \quad P_2 = \frac{P_1}{8} + \frac{T - 4T}{2r}$$

$$P_2 = \frac{P_1}{8} - \frac{3T}{2r} = 500$$

15. 1.40

Sol.

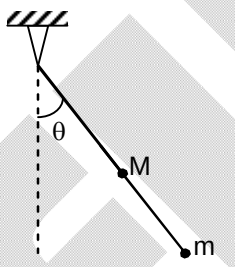


$$I_1 = \frac{14}{10} = 1.4$$

$$I_2 = I_3 = 0$$

16. 0.82

Sol. Angular acceleration of rod



$$\alpha = \frac{m(x+L)g\sin\theta}{m\left(x^2 + L^2\right)}$$

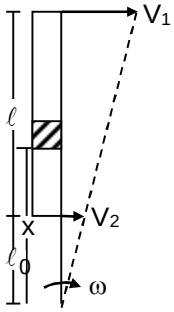
For rod to fall as fast as possible, $\dfrac{d\alpha}{dx} = 0$

or $\quad x = \left(\sqrt{2} - 1\right)L = 0.82$ m

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17. 3.50

Sol. $V_1 = \omega(\ell + \ell_0)$ & $V_2 = \omega\ell_0$



$$\therefore \quad \ell_0 = \frac{V_2\ell}{V_1 - V_2}$$

$$F = \int_{\ell_0}^{\ell} \frac{\eta\, b dx}{t} \frac{V_1 - V_2}{\ell} x$$

$$= \frac{\eta b \ell (V_1 + V_2)}{2t}$$

$$= 3.5$$

18. 3.00

Sol. Time taken by the particle to move from extreme position to half of amplitude is $\frac{T}{6}$.

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Chemistry

PART – II

Section – A

19. ABCD

Sol. (A) Calamine, $ZnCO_3$ is the ore of zinc.

(B) Proustite, Ag_3AsS_3 is the ore of silver.

(C) Cassiterite, SnO_2 is the ore of tin.

(D) Diaspore, $Al_2O_3.H_2O$ is the ore of aluminium.

20. AD

Sol. NaA is a salt of strong base and weak acid so its solution will be basic.

21. BC

Sol. Since H atoms are very small, so steric hindrance between them in eclipsed as well as staggered conformation is nearly same. So, torsional strain (i.e. repulsion between electron cloud of C–H bonds) and interaction between σ C–H (bonding) and σ^* C–H (antibonding) becomes main factors.

22. CD

Sol. In compound (I), the two benzene rings are in different planes due to steric repulsions of methyl groups and hence, (I) is not as reactive as (II) towards EAS.

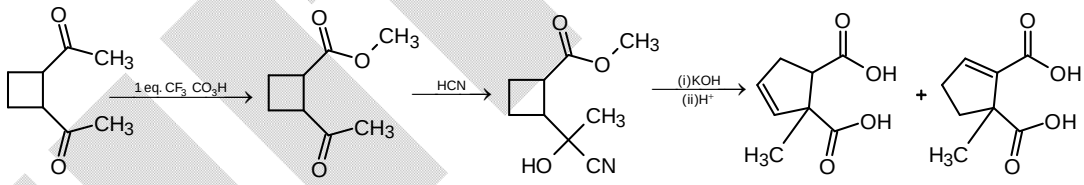
Methyl group is more stable at the equatorial position as compared to the axial position and hence, both the compounds are not formed in equal quantities.

When but-1-ene reacts with D_2, we get a chiral carbon. The product formed is a racemic mixture, which can be resolved.

During dehydration, carbocations are formed. The bridgehead carbocations are not stable, hence (V) is more reactive towards dehydration.

23. ABD

Sol.



24. ACD

Sol. (A) $Cl_2 + SeO_3^{2-} + H_2O \longrightarrow SeO_4^{2-} + 2Cl^- + 2H^+$

(B) $2FeSO_4 \longrightarrow Fe_2O_3 + SO_2 + SO_3$

(C) $Cl_2 + IO_3^- + 2OH^- \longrightarrow IO_4^- + 2Cl^- + H_2O$

(D) $Br_2 + 2OH^- \longrightarrow BrO^- + Br^- + H_2O$

25. D

Sol. Conceptual.

26. B

Sol. Conceptual.

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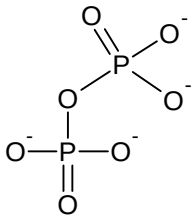
27. C
Sol. Conceptual.

28. B
Sol. Conceptual.

Section – B

29. 3.75
Sol. total possible geometrical isomers = 9 (3 are optically inactive and 6 are optically active)
Hence $3 \times 1 + 6 \times 2 = 15$ total stereoisomer

30. 1.50
Sol. as x= 6



has equivalent resonance in all terminal oxygens.

31. 4.20
Sol. $\frac{P_o - P_s}{P_o} = X_{solute}$

$\therefore \quad \frac{100 - 98.7}{100} = X_{solute} = 0.013$

Now $\Delta T_f = K_f \times m$

$0.72 = K_f \times \frac{0.013 \times 1000}{0.978 \times 78}$

$\therefore \quad K_f \approx 4.2 \text{ K Kg / mol}$

32. 1.33
Sol. The formula of the product formed when formaldehyde reacts with ammonia is $(CH_2)_6N_4$.
The name of the product is hexamethylene tetramine also called Urotropine.
$6HCHO+4NH_3 \rightarrow (CH_2)_6N_4+6H_2O$

33. 9.00
Sol. For unielectron species energy of 3s, 3p, 3d are equal. Hence total number of degenerate orbitals equal to 1+3+5 = 9

34. 42.22
Sol. $\left(P + \frac{an^2}{v^2}\right)(v - nb) = nRT$

$n = 1$

$\left(p + \frac{a}{v^2}\right)(v - b) = RT$

If b is negligible

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$$\left(p+\frac{a}{v^2}\right)(v)=RT$$

$$p=\frac{RT}{v}-\frac{a}{v^2}$$

The equation is quadratic in V thus

$$V=\frac{+RT\pm\sqrt{R^2T^2-4ap}}{2p}$$

Since V has one value at given P and T, thus numerical value of discriminant = 0

$R^2T^2 = 4ap$

$$p=\frac{R^2T^2}{4a}=\frac{(0.821)^2(300)^2}{4\times3.592}$$

$\therefore p = 8\times5.277$

p = 00042.22

35. 2.00

Sol. $3Br_2 + 6NaOH = 5NaBr + NaBrO_3$

$NaBr + AgNO_3 = AgBr + NaNO_3$

$AgBr + KMnO_4 = Br_2 + MnO_2$

No. of meq of $KMnO_4$ = No. of meq of AgBr

No. of millimoles of NaBr =2/3 x 3 x 50 = 100

no. of millimoles of NaOH required = 6/5 x100 = 120

volume of NaOH used = 60 ML

molarity of NaOH = 120/60 = 2M

36. 729.00

Sol. Temperature coefficient = 3 so an increasing 120^oC rate = 3^{12}

729 = 3^6 Hence the value of $\frac{x}{3^6} = 3^6 = 729$

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Mathematics

PART – III

Section – A

37. BC

Sol. Given

$$\frac{z_1}{z_2}+\frac{\bar{z}_1}{\bar{z}_2}=2 \Rightarrow \text{Re}\left(\frac{z_1}{z_2}\right)=1$$

Let $\frac{z_1}{z_2}=1+iy$

Now we have $\arg\left(\frac{z_1-z_2}{-z_2}\right)=\text{Arg}\left(1-\frac{z_1}{z_2}\right)$

$= \text{Arg}\,(-iy)=\pm\frac{\pi}{2}$

38. CD

Sol. Subtraction both equation weight

$$(a-b)(ab-a-b-2)(x-1)=0$$

weight $ab=a+b+2$

$b=1+\frac{b}{a}+\frac{2}{a}<3$ since a > 1, b > 1 and $a \neq b$

hence b = 2, a = 4

39. AB

Sol. Let the equation of other asymptote is $x+y+\lambda=0$,

tangent meet both the asymptote at $A(0,2)$ and $B(0,-\lambda)$

mid of both the point is P

Hence $\lambda=0$

Now area of triangle ABP is

$\frac{1}{2}\begin{vmatrix}0 & 0 & 1\\ 0 & 2 & 1\\ -1 & 1 & 1\end{vmatrix}=1$, hence latus rectum is 2.

40. AC

Sol. $f(x)=k\left(9^x-3^x\right)$

$f'(0)=k(\ln 9-\ln 3)=\ln 3$

k = 1

Hence $f(x) = 9^x - 3^x$

41. AC

Sol. $f(\alpha)=\frac{15}{1-\alpha}$

Hence $N=\frac{15^{14}}{(1-\alpha)\left(1-\alpha^2\right)\left(1-\alpha^3\right)-\left(1-\alpha^{14}\right)}=15^{13}$

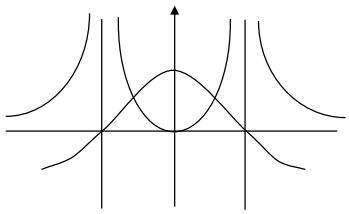
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42. AB

Sol. Equation of curve P is $2y = 1 + x^2$

43. B

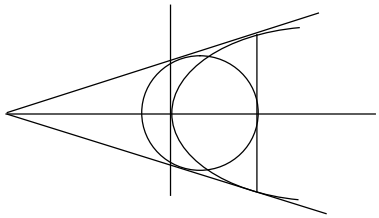
Sol. (I) Use graph. (Hence 4 point)



(II) $\underline{|7} = 2^4 \cdot 3^2 \cdot 5^1 \cdot 7^1$

(III) roots of $2x^2 + 3x + 4 = 0$ are imaginary

(IV) $r = 4\left(\sqrt{2}+1\right)$ or $r = 4\left(\sqrt{2}-1\right)$



44. A

Sol. $A^T(P+Q)A = \left(P^T - Q^T\right)\left(P^T + Q^T\right)^{-1}(P+Q)(P+Q)^{-1}(P-Q)$

$= (P+Q)(P-Q)^{-1}(P-Q) = P+Q$

45. B

Sol. (I) $x_1 + x_2 = 2m,\ x_1x_2 = m$ then

$8m^3 - 3m(2m) = 4m^2 - 2m \Rightarrow 8m^3 - 10m^2 + 2m = 0$

$m = 0, m = 1, m = \frac{1}{4}$

(II) $p = 4, \mu = 2$

(III) Let $x = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$

then $d = b + \frac{1}{2b}, a = \frac{1}{2b} - b, c = 2b$

Also, $b = \pm\frac{1}{\sqrt{6}}$, or $b = \pm\frac{1}{\sqrt{2}}$

(IV) $\det\left(A^2BC^{-1}\right) = \frac{|A|^2|B|}{|C|} = \frac{12}{5}$.

46. C

Sol. 6 either occurs at the 1st , 2nd or 5th place, so total number of ways are 8.9^3 and If it occurs at the 1st place the total number of ways are 9^4 so required number of ways are $9^3 \times 8 + 9^4$ $\lambda = 17$

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Section – B

47. 1.00

Sol. Put $z = x + iy$

$x^2 - y^2 + 2i \times y = x + iy + x^2 + y^2 + 2$

$-y^2 = x + y^2 + 2$ and $2 \times y = y$

$y = 0$ $x = 1/2$

$x = -2$ No

48. 49.00

Sol. Value of determinant $\left(\alpha^3 + \beta^3 + \gamma^3 - 3\alpha\beta\gamma\right)^2$

49. 7.00

Sol. Let missing number is

Also $\dfrac{\frac{n(n+1)}{2} - n}{n-1} \geq \dfrac{602}{17} \geq \dfrac{\frac{n(n+1)}{2}}{n-1}$

$n = 69, 70$

$x = 7,$ when $n - 69$

50. 8.00

Sol. Let common ratio is $\dfrac{n}{m}$ where n > m & m are co-prime then

Let $a_1 = km^4$ then

$a_5 = a_1 \dfrac{n^4}{m^4}$ $k \in I^+$

Also, $kn^4 < 100 \Rightarrow n \leq 3$

Hence total number of sets are 8.

51. 5.00

Sol. Let r be the roots of the equation then $r - 2 = \dfrac{1}{\pi} \Rightarrow r^2 - 2r - 1 = 0$

Hence $b = -2, c = -1$ hence $b^2 + c^2 = 5$

52. 1275.00

Sol. of x^{1274} is $\left(1^2 - 2^2 + 3^2 - 4^2 + ... + 49^2 - 50^2\right)$

53. 13.00

Sol. If one box has exactly α white balls and the other box has β white balls, then

$\dfrac{\alpha}{20} \cdot \dfrac{\beta}{20} = 0.21 \Rightarrow \alpha\beta = 84$

Since $\alpha, \beta \in I^+$

$(\alpha, \beta) \equiv (6,14), (7,12)$

hence $\alpha = 7, \beta = 12$

Then the required probability = $\dfrac{(20-7)(20-12)}{400} = 0.26$

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54. 17.00

Sol. Let $z = a + ib = (z + n)4i$

or $a = -4b$ and $b = 4a + 4n = 4(-4b + n)$

or $n = \frac{b}{4} + 4b = 1 + 16 = 17$

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